In [1]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

print("Libraries imported successfully.")
```

```
Libraries imported successfully.
```

In [2]:
```
pages = ["A", "B", "C", "D"]

links = {
    "A": ["B", "C"],
    "B": ["C"],
    "C": ["A"],
    "D": ["C"]
}

print("Pages:", pages)
print("Links:")

for page, destinations in links.items():
    print(page, "->", destinations)
```

```
Pages: ['A', 'B', 'C', 'D']
Links:
A -> ['B', 'C']
B -> ['C']
C -> ['A']
D -> ['C']
```

In [3]:
```
n = len(pages)

page_index = {
    page: i for i, page in enumerate(pages)
}

transition = np.zeros((n, n))

for source, destinations in links.items():
    if destinations:
        probability = 1 / len(destinations)

        for destination in destinations:
            transition[
                page_index[destination],
                page_index[source]
            ] = probability

print("Transition matrix:")

display(
    pd.DataFrame(
        transition,
        index=pages,
        columns=pages
```

```
    )
)
```

Transition matrix:

	A	B	C	D
A	0.0	0.0	1.0	0.0
B	0.5	0.0	0.0	0.0
C	0.5	1.0	0.0	1.0
D	0.0	0.0	0.0	0.0

In [4]:
```
damping = 0.85

rank = np.ones(n) / n

tolerance = 1e-8
max_iterations = 100

for iteration in range(max_iterations):
    new_rank = (
        (1 - damping) / n
        + damping * transition.dot(rank)
    )

    dangling_pages = [
        page for page in pages
        if len(links[page]) == 0
    ]

    if dangling_pages:
        dangling_rank = sum(
            rank[page_index[page]]
            for page in dangling_pages
        )

        new_rank += damping * dangling_rank / n

    if np.linalg.norm(new_rank - rank, ord=1) < tolerance:
        rank = new_rank
        break

    rank = new_rank

print("Iterations:", iteration + 1)
print("PageRank scores:")

results = pd.DataFrame({
    "Page": pages,
    "PageRank": rank
})

results = results.sort_values(
    "PageRank",
```

```
    ascending=False
).reset_index(drop=True)

display(results)
```

```
Iterations: 37
PageRank scores:
```

	Page	PageRank
0	C	0.394149
1	A	0.372527
2	B	0.195824
3	D	0.037500

In [5]:
```
plt.figure(figsize=(7, 4))

plt.bar(
    results["Page"],
    results["PageRank"]
)

plt.title("PageRank Scores")
plt.xlabel("Web Page")
plt.ylabel("PageRank Score")

plt.tight_layout()
plt.show()
```

